ELEC7023 Assignment 3–2

ELEC7023 Assignment 3–2 — Object Detection

Item	Content
Course	Edge computing device programming for AI projects
Student	Yao Yichuan
Task	Detect objects in **two images**, compare confidence thresholds **0.3** and **0.7**
Device	Jetson (jetson–inference `detectNet`)
Code	`assin3_detection.py`
Model	`ssd-mobilenet-v2`

GitHub Repository

https://github.com/yichuanyao/7023B–Assignment–Three

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1. Method

Test images (from COCO detection/segmentation benchmark):

Image	Content	Source
`images/image1.jpeg`	Two cats + remotes on a sofa (COCO `000000039769`)	COCO val2017
`images/image2.jpeg`	Kitchen scene with person & many objects (COCO	COCO val2017

Image	Content	Source
	`000000397133`)	

1. Load `ssd-mobilenet-v2` with `jetson_inference.detectNet`.
2. Change camera input to **image input** (`jetson_utils.loadImage`).
3. Run detection on **Image 1** and **Image 2**.
4. Repeat with threshold **0.3** (< 0.5) and **0.7** (> 0.5).
5. For each detection, record:
   `ClassID, Confidence, Left, Top, Right, Bottom, Width, Height, Area, Center`.
6. Save annotated images under `results/`.

Run command (on Jetson)

```
cd ~/jetson-inference/examples
python3 assin3_detection.py --image1 images/image1.jpeg --image2
        images/image2.jpeg --thresholds 0.3 0.7 --outdir results
```

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2. Detected Images

Image 1 @ threshold 0.3 (5 detections)

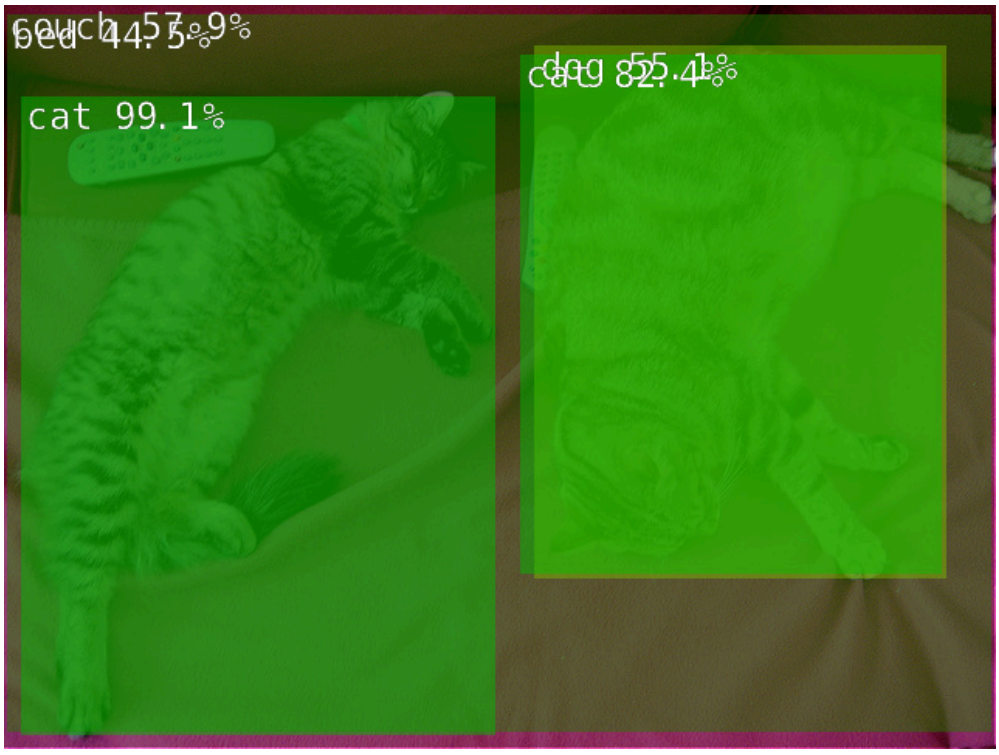


Image 1 threshold 0.3

Image 1 @ threshold 0.7 (2 detections)



Image 1 threshold 0.7

Image 2 @ threshold 0.3 (4 detections)

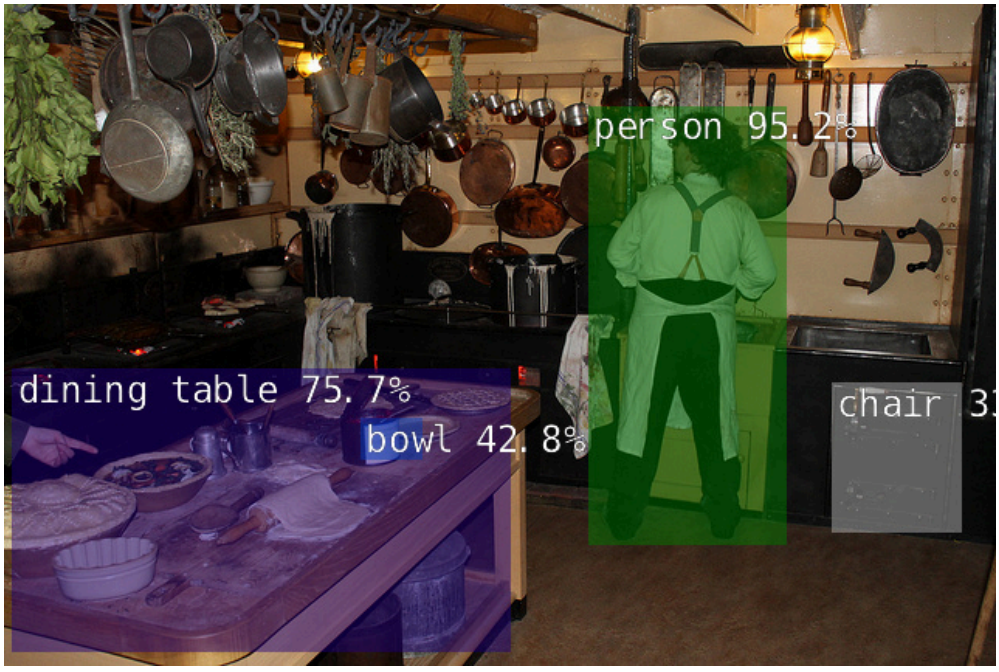


Image 2 threshold 0.3

Image 2 @ threshold 0.7 (2 detections)

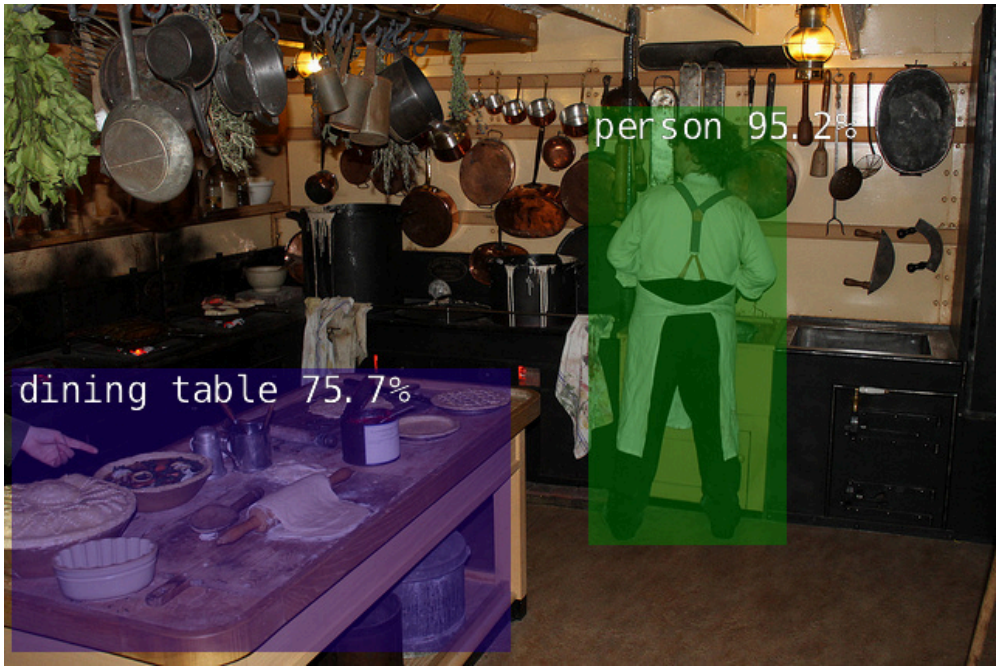


Image 2 threshold 0.7

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3. Detection Results

Image 1 @ threshold 0.3 (5 objects)

```
-- Detection #1
-- ClassID: 63
-- Class: couch
-- Confidence: 0.57861328125
-- Left:    0.0
-- Top:     0.0
-- Right:   639.0
-- Bottom:  479.0
-- Width:   639.0
-- Height:  479.0
-- Area:    306081.0
-- Center:  (319.5, 239.5)

-- Detection #2
-- ClassID: 65
-- Class: bed
-- Confidence: 0.444580078125
-- Left:    2.65625
-- Top:     6.5625
-- Right:   638.125
-- Bottom:  470.15625
-- Width:   635.46875
-- Height:  463.59375
-- Area:    294599.34375
-- Center:  (320.390625, 238.359375)
```

```
-- Detection #3
-- ClassID: 17
-- Class: cat
-- Confidence: 0.99072265625
-- Left:    11.484375
-- Top:     59.296875
-- Right:   318.125
-- Bottom:  471.328125
-- Width:   306.640625
-- Height:  412.03125
-- Area:    126345.5234375
-- Center:  (164.8046875, 265.3125)

-- Detection #4
-- ClassID: 18
-- Class: dog
-- Confidence: 0.55078125
-- Left:    342.5
-- Top:     26.71875
-- Right:   609.375
-- Bottom:  371.25
-- Width:   266.875
-- Height:  344.53125
-- Area:    91946.78125
-- Center:  (475.9375, 198.984375)

-- Detection #5
-- ClassID: 17
-- Class: cat
-- Confidence: 0.82421875
-- Left:    333.125
-- Top:     32.9296875
-- Right:   605.625
-- Bottom:  367.96875
-- Width:   272.5
-- Height:  335.0390625
-- Area:    91298.140625
-- Center:  (469.375, 200.44921875)
```

Image 2 @ threshold 0.3 (4 objects)

```
-- Detection #1
-- ClassID: 67
-- Class: dining table
-- Confidence: 0.7568359375
-- Left:    5.3125
-- Top:     235.39208984375
-- Right:   328.125
-- Bottom:  418.86865234375
-- Width:   322.8125
-- Height:  183.4765625
-- Area:    59228.52734375
-- Center:  (166.71875, 327.13037109375)

-- Detection #2
-- ClassID: 1
-- Class: person
-- Confidence: 0.9521484375
-- Left:    377.8125
```

```
-- Top:     66.51025390625
-- Right:   505.9375
-- Bottom:  349.64794921875
-- Width:   128.125
-- Height:  283.1376953125
-- Area:    36277.015625
-- Center:  (441.875, 208.0791015625)

-- Detection #3
-- ClassID: 62
-- Class: chair
-- Confidence: 0.338134765625
-- Left:    534.6875
-- Top:     244.14892578125
-- Right:   619.0625
-- Bottom:  341.30810546875
-- Width:   84.375
-- Height:  97.1591796875
-- Area:    8197.8056640625
-- Center:  (576.875, 292.728515625)

-- Detection #4
-- ClassID: 51
-- Class: bowl
-- Confidence: 0.427734375
-- Left:    230.0
-- Top:     267.50048828125
-- Right:   270.0
-- Bottom:  295.43896484375
-- Width:   40.0
-- Height:  27.9384765625
-- Area:    1117.5390625
-- Center:  (250.0, 281.4697265625)
```

Image 1 @ threshold 0.7 (2 objects)

```
-- Detection #1
-- ClassID: 17
-- Class: cat
-- Confidence: 0.99072265625
-- Left:    11.484375
-- Top:     59.296875
-- Right:   318.125
-- Bottom:  471.328125
-- Width:   306.640625
-- Height:  412.03125
-- Area:    126345.5234375
-- Center:  (164.8046875, 265.3125)

-- Detection #2
-- ClassID: 17
-- Class: cat
-- Confidence: 0.82421875
-- Left:    333.125
-- Top:     32.9296875
-- Right:   605.625
-- Bottom:  367.96875
-- Width:   272.5
-- Height:  335.0390625
```

```
-- Area:    91298.140625
-- Center:  (469.375, 200.44921875)
```

Image 2 @ threshold 0.7 (2 objects)

```
-- Detection #1
-- ClassID: 67
-- Class: dining table
-- Confidence: 0.7568359375
-- Left:    5.3125
-- Top:     235.39208984375
-- Right:   328.125
-- Bottom:  418.86865234375
-- Width:   322.8125
-- Height:  183.4765625
-- Area:    59228.52734375
-- Center:  (166.71875, 327.13037109375)

-- Detection #2
-- ClassID: 1
-- Class: person
-- Confidence: 0.9521484375
-- Left:    377.8125
-- Top:     66.51025390625
-- Right:   505.9375
-- Bottom:  349.64794921875
-- Width:   128.125
-- Height:  283.1376953125
-- Area:    36277.015625
-- Center:  (441.875, 208.0791015625)
```

Field meanings (Lecture 4)

Field	Meaning
ClassID	Predicted object category ID
Confidence	How certain the model is
Left / Top	Top–left corner of the box
Right / Bottom	Bottom–right corner of the box
Width / Height	Box size
Area	Width × Height
Center	Midpoint of the box

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4. Threshold Comparison Table

Image	Confidence threshold	Number of detected objects	Missed or incorrect detections
Image 1	0.3	5	Incorrect: `bed` (sofa misclassified), `dog` (right cat misclassified as dog). Missed: two remotes
Image 1	0.7	2	Incorrect low-confidence boxes removed. Still missed: two remotes
Image 2	0.3	4	Weak extras: `chair` (0.34), `bowl` (0.43). Missed many kitchen items (pots/pans/utensils)
Image 2	0.7	2	`chair` and `bowl` filtered out. Still missed many small kitchen objects

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5. Report Questions

Q1. How did the results change when you increased the threshold?

When the threshold increased from **0.3** to **0.7**, the number of detections decreased:

- Image 1: **5 → 2**
- Image 2: **4 → 2**

Only high-confidence boxes remained (both cats on Image 1; person + dining table on Image 2). Low-confidence / uncertain predictions were removed, so the output looked cleaner, but some true or borderline objects disappeared.

Q2. Were any objects detected at the lower threshold but not at the higher threshold? Why?

Yes.

- Image 1 lost: `couch` (0.58), `bed` (0.44), `dog` (0.55)
- Image 2 lost: `chair` (0.34), `bowl` (0.43)

These scores are **between 0.3 and 0.7**, so they pass the low threshold but fail the high threshold. The network still proposes those boxes; confidence filtering discards them when the cutoff is stricter.

Q3. Which threshold would you choose for your application scenario? Explain your choice.

Chosen scenario: home / kitchen object detection on an edge device (person, cat, bowl, furniture — COCO-style classes).

Chosen threshold: 0.7

Reason: at 0.3, Image 1 produced clear mistakes (`bed`, `dog`), and Image 2 added weak boxes. At 0.7, the remaining detections (`cat`, `cat`, `person`, `dining table`) are more reliable. For an edge demo where false alarms are more confusing than missing a few small objects, **0.7** is the better choice.

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6. Submission Checklist

- ☐ `assin3_detection.py` uploaded to GitHub
- ☐ GitHub link written in this report
- ☐ Detected / annotated images included
- ☐ Detection field outputs included
- ☐ Comparison table filled
- ☐ Three questions answered
- ☐ Export this file to **PDF or DOC** and upload to **Moodle**